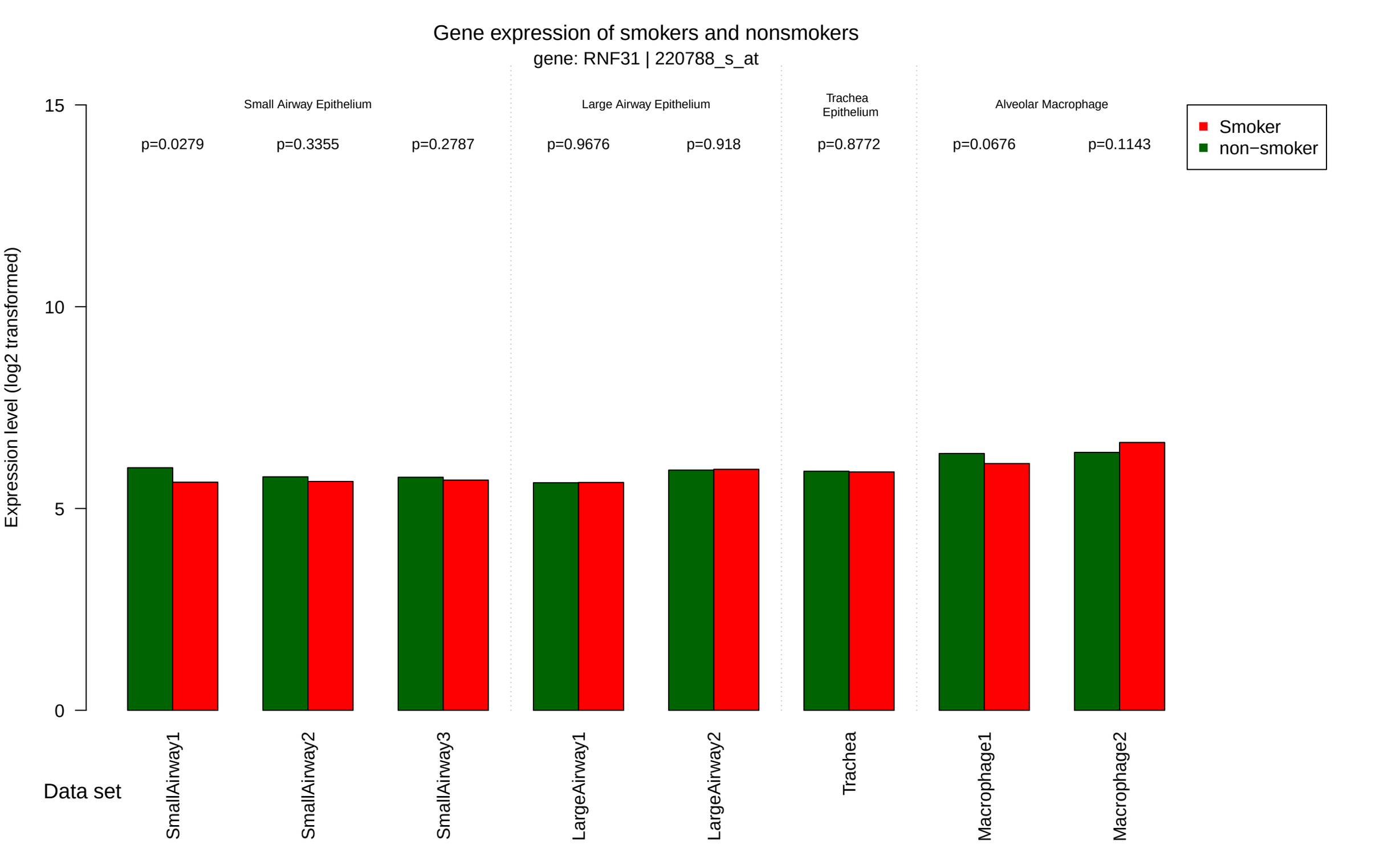
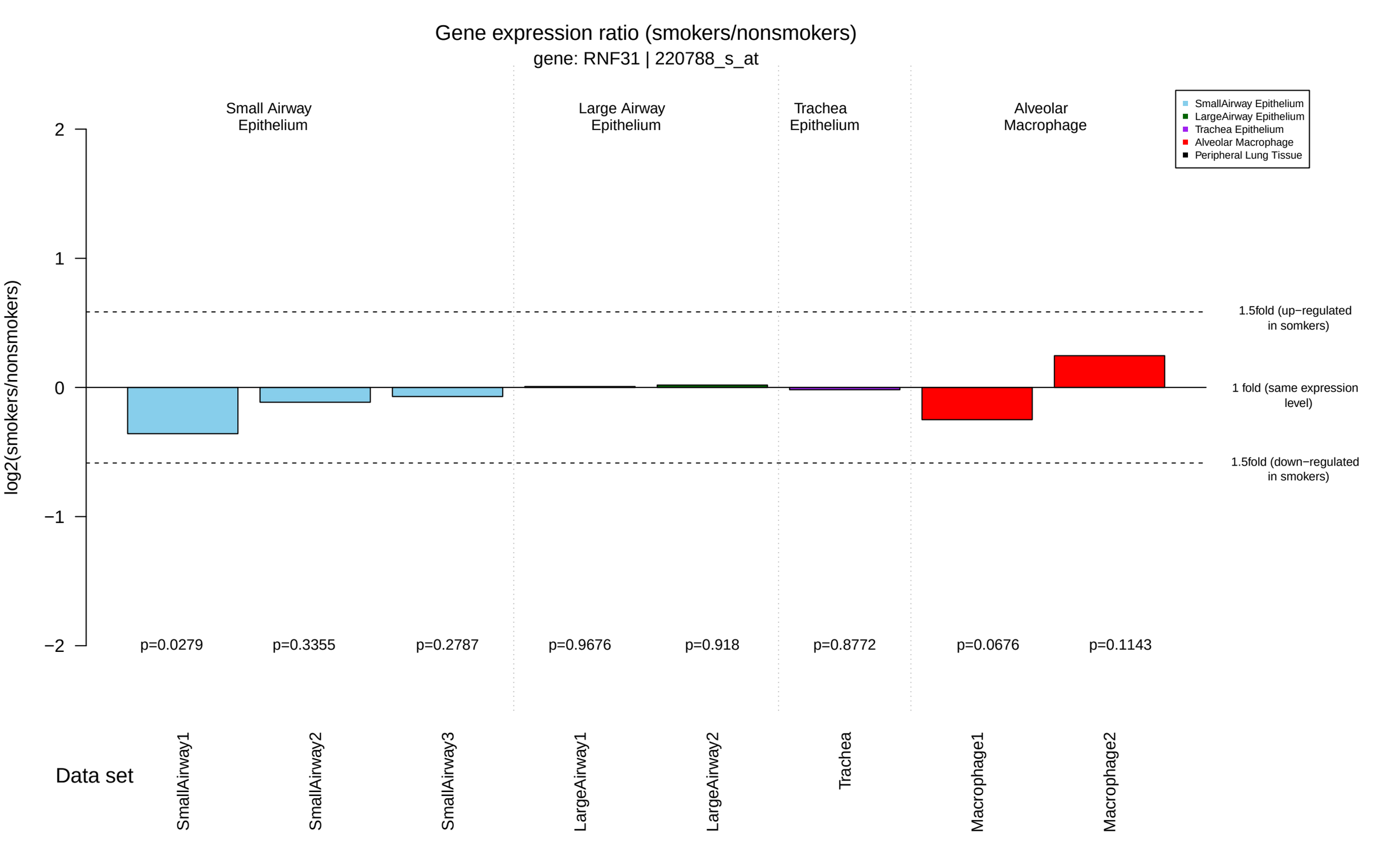
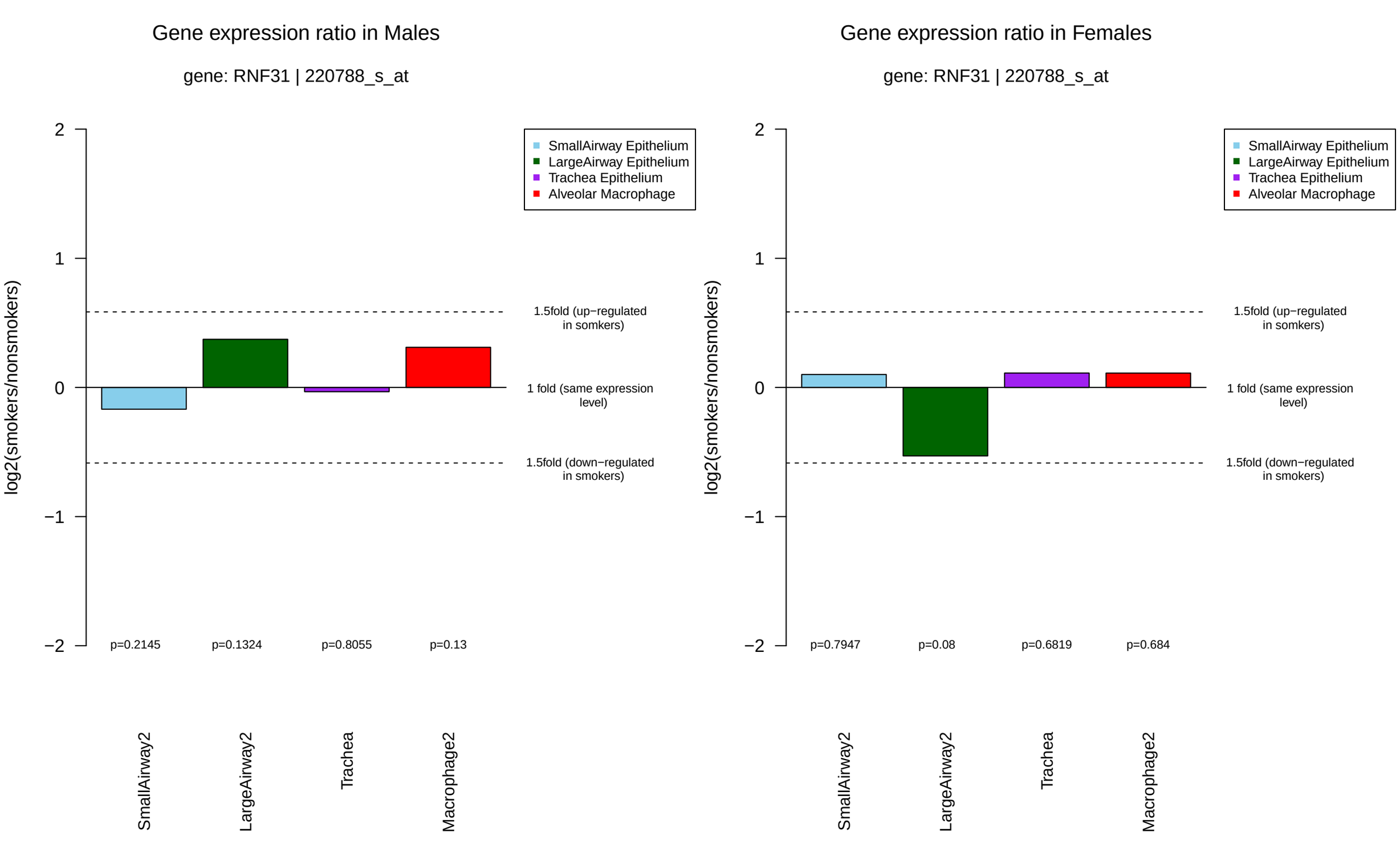
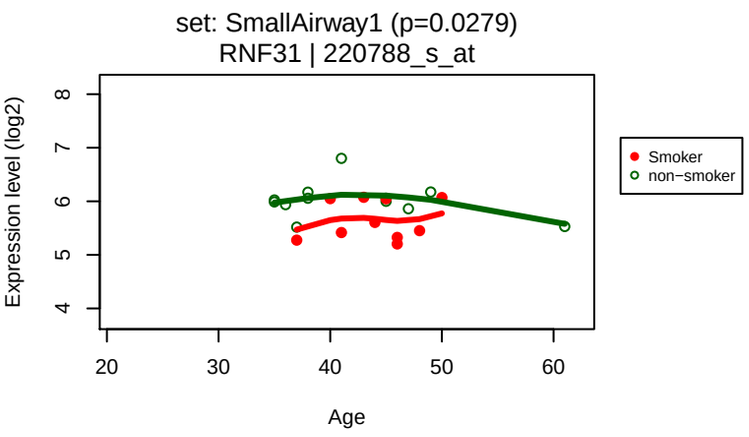
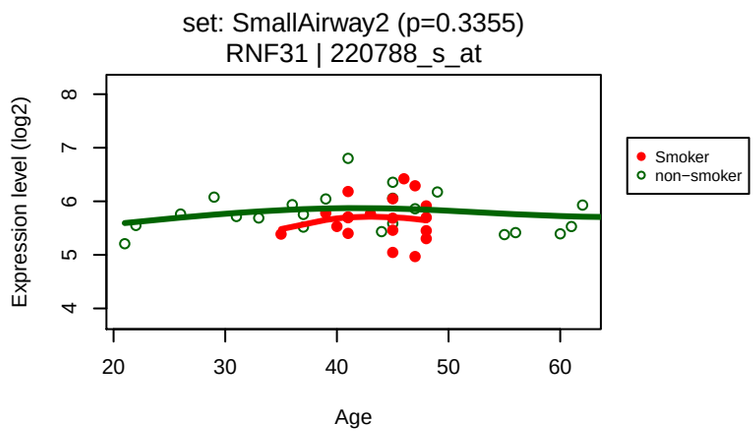
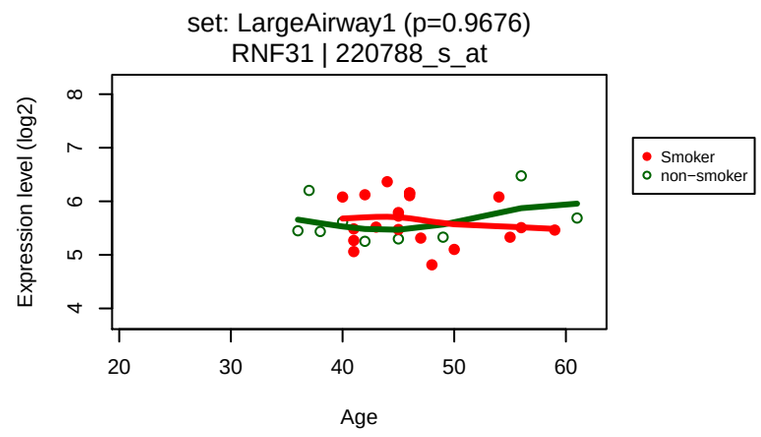

Gene expression of smokers and nonsmokers
gene: RNF31 | 220788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0279
p=0.3355
p=0.2787
p=0.9676
p=0.918
p=0.8772
p=0.0676
p=0.1143
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

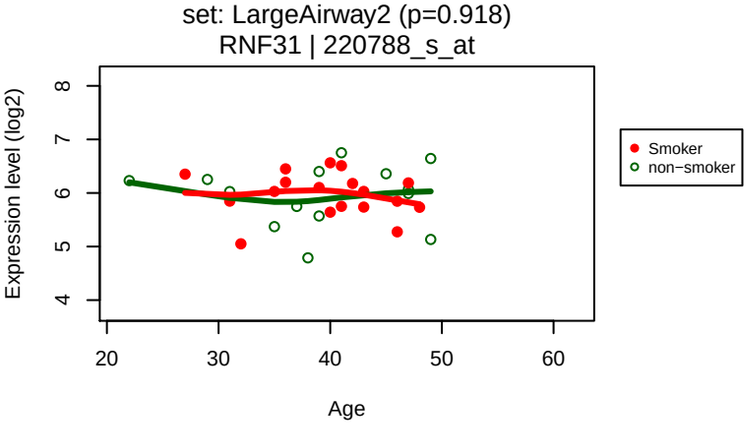

Gene expression ratio (smokers/nonsmokers)
gene: RNF31 | 220788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0279
p=0.3355
p=0.2787
p=0.9676
p=0.918
p=0.8772
p=0.0676
p=0.1143
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

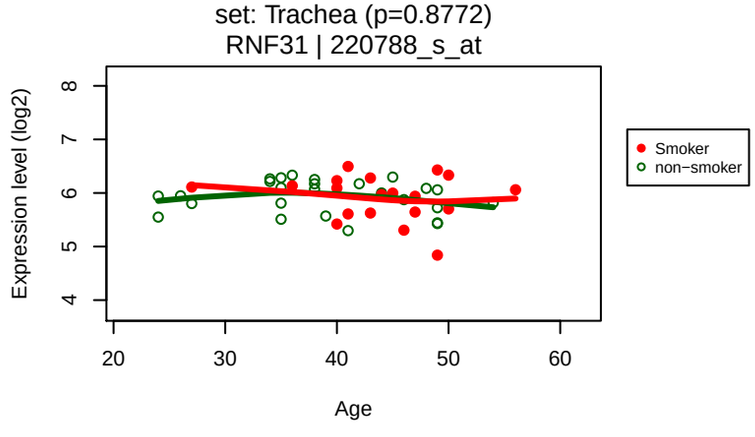

Gene expression ratio in Males
gene: RNF31 | 220788_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2145
p=0.1324
p=0.8055
p=0.13
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RNF31 | 220788_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7947
p=0.08
p=0.6819
p=0.684
SmallAirway2
LargeAirway2
Trachea
Macrophage2

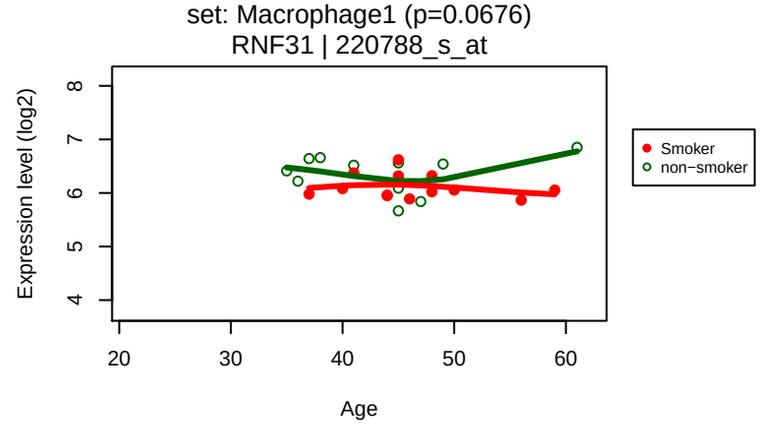

set: SmallAirway1 (p=0.0279)
RNF31 | 220788_s_at
Expression level (log2)
Age
Smoker
non−smoker

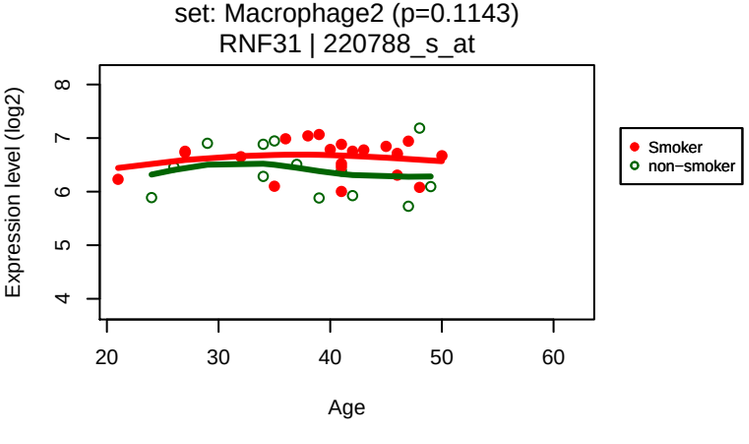

set: SmallAirway2 (p=0.3355)
RNF31 | 220788_s_at
Expression level (log2)
Age
Smoker
non−smoker

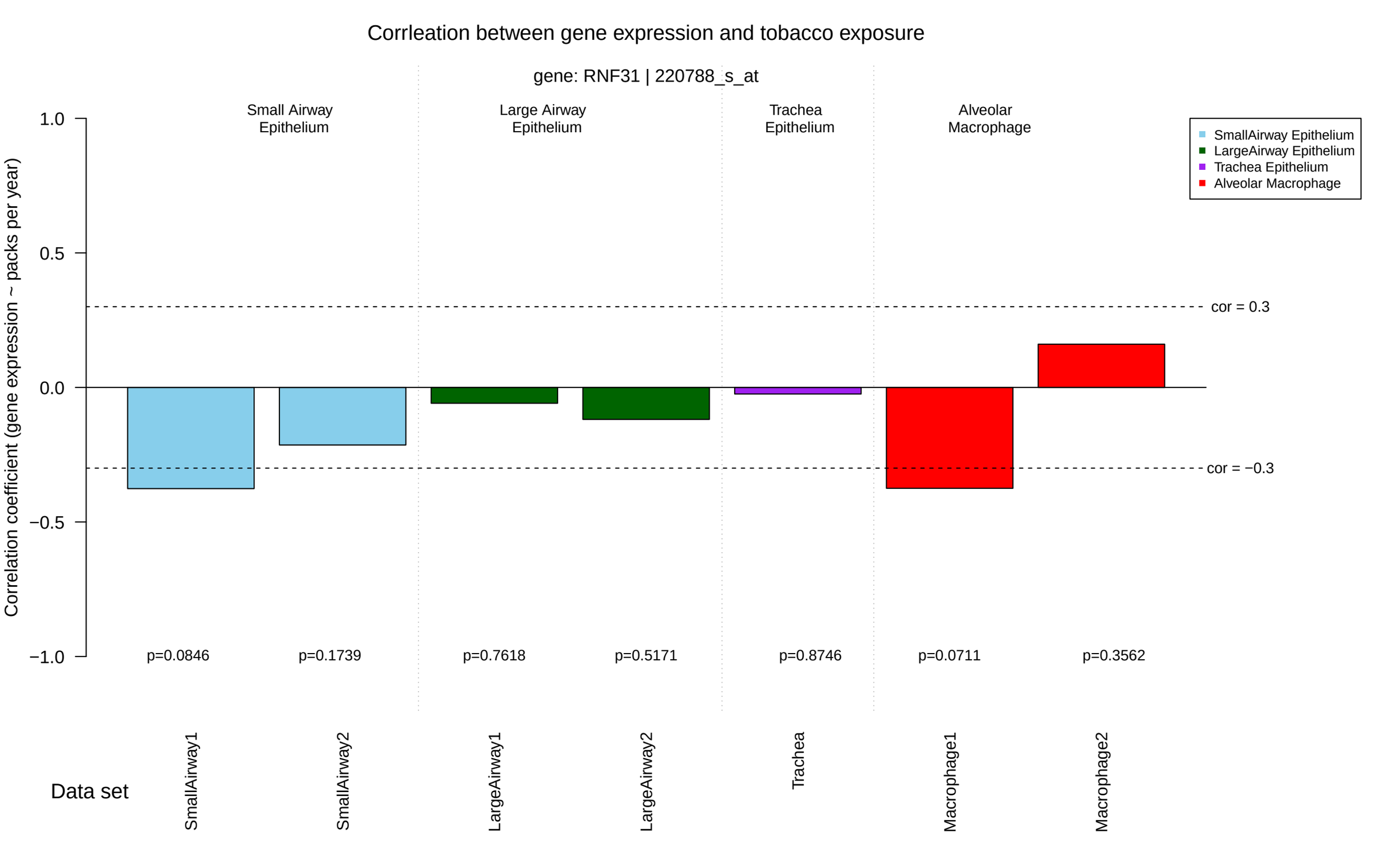

set: LargeAirway1 (p=0.9676)
RNF31 | 220788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.918)
RNF31 | 220788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8772)
RNF31 | 220788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0676)
RNF31 | 220788_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1143)
RNF31 | 220788_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RNF31 | 220788_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0846
p=0.1739
p=0.7618
p=0.5171
p=0.8746
p=0.0711
p=0.3562
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2